

The density of ethanol varies slightly with temperature, as shown in the table below.

Temperature (°C)	Density (g/mL)
4	0.803
8	0.800
12	0.796
16	0.793
20	0.789
24	0.786
28	0.782
32	0.779
36	0.776
40	0.772

Based on this data, how many molecules of ethanol are contained in 30 mL of ethanol at 8 °C? (Note: The molar mass of ethanol is 46.08 g/mol.)

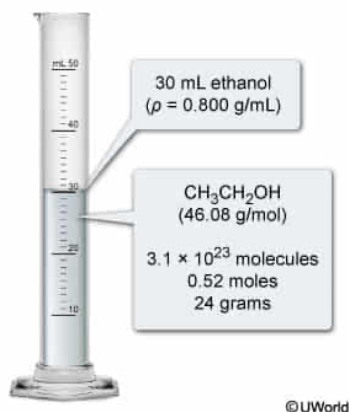
- ☐ A. 8.6×10^{-25} molecules [3%]
- ✓ ☒ B. 3.1×10^{23} molecules [78%]
- ☐ C. 4.9×10^{23} molecules [12%]
- ☐ D. 6.7×10^{26} molecules [5%]

Correct

78% answered correctly

Explanation:

Density, mass, and molecules of 30 mL ethanol at 8 °C



The **density** ρ of a substance is defined as the **mass** m it contains per unit of **volume** V that it occupies. This physical property is expressed mathematically as:

$$\rho = \frac{m}{V}$$

Because materials tend to expand when heated and contract when cooled, the volume that a substance occupies can undergo minute changes with temperature. As a result, the exact density of a substance **varies slightly with temperature**.

When the density of a substance at a given temperature is known, density can be used as a conversion factor to convert between the mass and volume of that substance. For example, 30 mL of ethanol at 8 °C has a density of 0.800 g/mL and a mass of 24 g.

30

mL

The **molar mass** of a compound is the amount of **mass contained in 1 mole** of the compound; it can be used to convert the mass of a compound to the number of moles. As such, dividing the mass of ethanol by its molar mass gives the number of moles of ethanol.

24

g

A **mole** is defined as **Avogadro's number** ($N_A = 6.022 \times 10^{23}$) of atoms or molecules. Accordingly, the number of molecules of ethanol is found by multiplying the moles of ethanol by N_A .

0.52

Therefore, 30 mL of ethanol at 8 °C contains 3.1×10^{23} molecules of ethanol.

(Choice A) 8.6×10^{-25} results from dividing rather than multiplying the moles of ethanol by N_A .

(Choice C) 4.9×10^{23} results from dividing rather than multiplying the volume of ethanol by the density.

(Choice D) 6.7×10^{26} results from multiplying rather than dividing the mass of ethanol by the molar mass.

Educational objective:

The density of a substance varies slightly with temperature and is defined as the mass of the substance per unit of volume that it occupies. Avogadro's number ($N_A = 6.022 \times 10^{23}$) of formula units equals one mole, and the molar mass of a compound is the amount of mass (in grams) contained in 1 mole of formula units (grams/mole).

Time spent: 0 sec

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System:
Atomic Theory and Chemical Composition